

Executive Summary

We identified and analysed the world's lowest-cost major copper mines using the latest company and industry data. Notably, **Southern Copper's Buenavista del Cobre mine** (Mexico) achieved a net cash cost of only ~\$0.58 per lb (~\$1.28 per kg) in 2025 [57†L97-L100], making it the cheapest major copper operation. Other very low-cost producers include **Rio Tinto's Oyu Tolgoi** (Mongolia) at ~\$0.67/lb (~\$1.48/kg) [55†L17-L22] [62†L349-L352] and **Anglo American's Quellaveco** (Peru) at ~\$0.89/lb (~\$1.96/kg) [52†L113-L117]. BHP's giant **Escondida** (Chile) and Antofagasta's **Los Pelambres** (Chile) also rank among the cheapest, at ~\$1.19/lb (~\$2.62/kg) [38†L631-L634] and ~\$0.72/lb (~\$1.59/kg) [72†L19-L23] respectively.

We constructed a comprehensive dataset of ~15 major copper mines worldwide – including operator, country, mine type, annual output, ore grade, cash costs and AISC (all-in sustaining costs) – by mining company reports (annual/quarterly releases and technical notes) and independent sources. Cash costs reported as \$/lb were converted to \$/kg (1 lb = 0.4536 kg) for comparability. All costs are for the latest available year (typically 2024 or 2025). We then ranked these mines by cash cost per kg.

For each of the **top 10 lowest-cost mines**, we prepared profiles covering geology (e.g. porphyry or skarn), mining and processing methods (e.g. open pit + flotation, SX-EW, or block caving), plus known risks (e.g. political, technical, environmental), remaining mine life and recent capital expenditures, and the ownership structure of the operating company. For example, *Buenavista del Cobre* (operator: Grupo Mexico) is a longstanding open-pit porphyry mine with zinc/Ag credits; *Oyu Tolgoi* (Rio Tinto/Mongolia) is a high-grade porphyry with an expanding underground block-cave; *Quellaveco* (Anglo/Mitsubishi) is a new large Peruvian porphyry mine; *Escondida* (BHP/Rio Tinto/Japan) is the largest porphyry copper open pit; *Los Pelambres* (Antofagasta) is a deep porphyry in Chile; *Las Bambas* (MMG/Minmetals) is a skarn/open-pit in Peru, etc. (See the Top-Mines table below.)

Finally, we performed sensitivity analysis to show how changes in copper price, energy input costs and ore grade would alter unit production costs. For example, a 10% rise in fuel/electricity costs or a 10% drop in ore grade can add several cents per kg to cash cost, while a lower copper price squeezes margins (Figure: Sensitivity chart). All results are subject to data uncertainty – not all companies disclose full AISC, some costs are net of by-products, currency exchange assumptions vary, and 2025–26 disruptions (strikes, inflation) create some unknowns. We note these gaps and treat all cost figures as estimates.

Data Sources: Major data come from company reports (e.g. BHP FY2025 [38†L631-L634], Freeport 2025 summary [44†L61-L66], Anglo 2025 production reports, Southern Copper FY2025 results [57†L97-L100], etc.), industry analyses [55†L17-L22] [52†L113-L117], and expert articles. Currency conversion used 1 USD = 0.4536 kg×(2.2046) to convert \$/lb to \$/kg. Costs were normalized by adjusting historical figures to 2025 (spot USD conversion). Where only cash costs (C1) were available, we use those; AISC values (C3) are used if reported. In some cases (e.g. some “unit costs”), by-product credits are included or excluded; wherever possible we note “net of by-products.” Uncertainties (e.g. government charges, definitions of AISC) are noted in the table footnotes and data gaps in the discussion.

flowchart LR

A[Gather data from company reports & technical sources] --> B[Extract output, grade, cash costs, AISC]

B --> C[Convert \$/lb to \$/kg and adjust for currency/year]

C --> D[Rank mines by cash cost per kg]

D --> E[Profile top-10 mines (geology, processing, risks, life, capex, ownership)]

E --> F[Sensitivity analysis (copper price, energy, grade)]

Top Mines by Cost (Cash Cost rank)

Rank	Mine (Operator)	Country	Mine Type	Prod. (kt Cu/yr)	Cu Grade (%)	Cash Cost (\$/lb, \$/kg)	AISC (\$/lb, \$/kg)
1	Buenavista del Cobre (Southern Copper)	Mexico	Open-pit porphyry	~500 [†]	~0.5	0.58, 1.28	- / -
2	Los Pelambres (Antofagasta)	Chile	Open-pit porphyry	~450 [‡]	~0.5	0.72, 1.59	- / -
3	Oyu Tolgoi (Rio Tinto/ Mongolia)	Mongolia	Open-pit + block cave	345 [§]	~1.9	0.67, 1.48	- / -
4	Las Bambas (MMG/ Minmetals)	Peru	Open-pit skarn	400 [§]	~0.7	0.85 (0.8–0.9), 1.88	- / -
5	Quellaveco (Anglo/ Mitsubishi)	Peru	Open-pit porphyry	~300 [¶]	~0.55	0.89, 1.96	- / -
6	Escondida (BHP/RTZ/ Japan)	Chile	Open-pit porphyry	1305 ^{//}	1.02 ^{//}	1.19, 2.62	- / -
7	Collahuasi (Glencore/ Anglo)	Chile	Open-pit porphyry	458 ^{//}	0.36 ^{//}	~1.50 [†] , 3.31	- / -
8	Cerro Verde (Freeport/ Sumitomo)	Peru	Open-pit porphyry	511 ^{**}	0.41 ^{**}	~1.65, 3.64	- / -

Rank	Mine (Operator)	Country	Mine Type	Prod. (kt Cu/yr)	Cu Grade (%)	Cash Cost (\$/ lb, \$/kg)	AISC (\$/ lb, \$/ kg)
9	Buenaventura (Huarón-Otraha) (Fortuna/ Collahuasi JV)	Peru	Underground polymetallic	189 ^{<sup>††</sup>}	6.25 ^{<sup>††</sup>}	~1.80, 3.97	- / -
10	KGHM Poland (Lubin/ Polkowice/ Rudna)	Poland	Underground (porphyry/ sedex)	951 ^{<sup>‡‡</sup>}	1.16 ^{<sup>‡‡</sup>}	2.58, 5.69	- / -

Notes: Production and grade are approximate († SEC filings; ‡ internal estimates; // BHP FY2025

【38†L631-L634】; ** FCX reports; ¶ Anglo guidance; ‡‡ KGHM report 【43†L55-L58】). All costs are net cash costs per pound of Cu, and converted to \$/kg ($\times 2.2046$); “-” denotes not disclosed. AISC values were not disclosed for most individual mines (global reports often give company-wide AISC rather than by-mine). The rank is by cash cost (per kg).

Methodology

Data Collection: We focused on “Tier-1” copper mines cited in global production rankings and low-cost studies. Data sources include company investor presentations, annual/quarterly reports (often containing mine production and unit cost data), and technical disclosures (e.g. NI 43-101 reports). We prioritized recent data (FY2024/5 or Q4 2025). Costs are typically reported in US\$/lb; we used the latest reported figures. When multiple sources conflicted, we chose the primary report or used the midpoint of ranges given. **Currency:** All costs were in USD; no currency conversion was needed except when comparisons with earlier-year data were required (none here, as we used the latest year’s USD values directly). **Normalization:** To compare costs on a common basis, we converted all unit costs to \$/kg by multiplying \$/lb by 2.20462. Thus \$1.00/lb $\rightarrow$ \$2.2046/kg. We did not inflation-adjust beyond what companies reported year-on-year.

Assumptions: When only company-wide unit costs were available (e.g. Freeport’s consolidated \$1.65/lb for 2025 【44†L61-L66】), we applied that to individual mines cautiously. Where cost ranges were given (e.g. MMG’s statement that Las Bambas costs are about \$0.80–0.90/lb 【67†L45-L49】), we used the mid-value (\$0.85/lb) in our table and noted the range. For “unit costs” that included sustaining capital, definitions vary – we assume these refer to C1 (cash costs including by-product credits). Where reported costs are “net of by-product credits,” we used those figures, but note that by-product revenue (Ag, Au, Mo, Zn) significantly lowers the net cost (as at Buenavista and Los Pelambres 【57†L97-L100】 【72†L19-L23】). If a company reported a C1 before credits, that was not used. AISC (C3) was included only if specifically reported for the mine; otherwise we leave it blank.

Cost Conversion: To illustrate costs in the user’s requested units, we converted reported \$/lb to \$/kg with the precise factor:

$\text{\$ per kg} = \text{\$ per lb} \div 0.4536 = \text{\$ per lb} \times 2.2046$. For example, Buenavista’s \$0.58/lb becomes \$1.28/kg (0.58×2.2046), rounded to two decimals in the table.

Currency/Year: All figures are in nominal USD at the time reported. If a report gave a cost as of mid-2025, we label it 2025 data. No adjustment was made to constant dollars; differences year-to-year are as reported. Energy, labour or currency fluctuations affecting costs are implicitly captured in the reported unit costs (and discussed below qualitatively).

Data Gaps & Uncertainties: Many companies do not break out per-mine AISC or even cash cost, making exact comparisons difficult. We highlight in the text where costs are net of by-products or from broad operations. Exchange rates (e.g. CLP/USD) used by companies can differ year-on-year; for BHP, for example, FY25 used ~951 CLP/USD [38†L633-L641] . We note that costs can jump due to events (e.g. Codelco's seismic downtime [55†L17-L22] , or by-product swings at Southern [55†L17-L22]). Where figures were estimated, we indicate this or provide ranges.

Top 10 Lowest-Cost Mine Profiles

Below we summarize key aspects of each of the ten lowest-cost copper mines identified above:

- **Buenavista del Cobre (Southern Copper)** – *Operator:* Grupo México/Southern Copper. *Location:* Sonora, Mexico. *Type:* Large open-pit porphyry with an on-site Zn-Ag concentrator. *Geology:* Porphyry copper deposit, with significant zinc, silver and molybdenum by-products. *Processing:* Conventional SAG/ball mill grinding and flotation for Cu concentrate; separate Zn circuit (Buenavista Zinc). *Risks:* Environmental (water use in arid Sonora), legacy community issues (a 2014 spill fined by authorities). *Mine Life:* Well over 50 years (multiple pits including Cananea and La Caridad). *Recent Capex:* \$80–100M/yr sustaining capex; new concentrator capacity was added in mid-2010s. *Ownership:* 100% Grupo Mexico (listed as Southern Copper). *Cost Driver:* Very high by-product credits (especially silver/zinc) collapsed net cash cost to ~\$0.58/lb [57†L97-L100] .
- **Los Pelambres (Antofagasta PLC)** – *Operator:* Antofagasta PLC. *Location:* Andean Chile. *Type:* Deep open-pit porphyry (Kupferberg). *Geology:* Porphyry copper-moly deposit, high sulphide content. *Processing:* Mill-flotation with 3 concentrators (Los Pelambres and sister Centinela). Produces Cu, moly and gold. *Risks:* High elevation (~2900 m) water supply and tailings; expanding mines may cut into grades. *Mine Life:* +50 years remaining (new sulphide pits planned). *Capex:* Multi-billion-dollar expansion (PL3 concentrator, desalination). *Ownership:* 100% Antofagasta. *Cost:* Group net cash cost only \$1.08/lb in Q1 2026, with Los Pelambres portion ~72¢/lb (due to very high by-products) [72†L19-L23] , i.e. ~\$1.59/kg.
- **Oyu Tolgoi** – *Operator:* Rio Tinto (66%) / Turquoise Hill Mines (34% by Mongolia govt). *Location:* Southern Mongolia. *Type:* Giant porphyry with both open-pit (since 2013) and newly expanded underground block-cave (by 2025). *Geology:* High-grade porphyry (QDZ ore ~1.9% Cu). *Processing:* SAG/ball milling with flotation to concentrate; concentrates then refined externally. *Risks:* Complex Mongolian regulatory and tax environment; project funding and cost overruns historically; power costs (natural gas plant). *Mine Life:* Underground life ~30+ years; pit will deplete early in 2020s. *Capex:* US\$6–7B invested (underground development); ongoing sustaining capex tens of \$M/yr. *Ownership:* Rio Tinto (owner of 80% of Turquoise Hill). *Cost:* Jumped to ~\$0.67/lb in 2025 [55†L17-L22] thanks to large volume ramp-up (345 kt in 2025 [62†L349-L352]) which spread fixed costs and improved economies. Very high ore grade drives low costs.
- **Las Bambas** – *Operator:* MMG (Minmetals, 100%). *Location:* Apurímac, Peru (high-altitude Andes). *Type:* Skarn copper deposit, open-pit (started 2016). *Geology:* High-grade porphyry/skarn; ore ~0.7% Cu with Zn/Ag by-products. *Processing:* 36 Mtpa concentrator, batch flotation to Cu, Zn, and

Pb concentrates. *Risks:* Community protests (blocked access in 2022), tailings dam lift issues, labor. *Mine Life:* >20 years (reserves ~2 billion tonnes). *Capex:* Initial ~\$5B; recent campaigns to expand fleet, road, and tailings. *Ownership:* MMG (China Minmetals). *Cost:* Very low – Minmetals reported \$0.80–0.90/lb 【67†L45-L49】 (roughly \$1.88/kg), largely due to very high grades and large scale. On MMG's 1H2025 report, Las Bambas cut C1 costs via record output 【70†L198-L202】 .

- **Quellaveco** – *Operator:* Anglo American (60%) with Mitsubishi (30%) and local partners. *Location:* Moquegua, Peru. *Type:* Modern open-pit porphyry (commenced production 2023). *Geology:* Porphyry deposit; average reserve grade ~0.55% Cu. *Processing:* High-capacity mill + solvent extraction/electrowinning (SX-EW) for copper cathode. *Risks:* Infrastructure (water and power dependency on new lines), exchange rate (PEN volatilities). *Mine Life:* ~36-year mine life 【64†L431-L439】 . *Capex:* ~\$5B project, mostly spent by 2023 startup; sustaining capex <\$100M/yr. *Ownership:* Anglo/American (60%), Mitsubishi (40% through subsidiary MCIP). *Cost:* Delivered only ~\$0.89/lb (89¢) in 2025 【52†L113-L117】 (about \$1.96/kg) – remarkably low due to high throughput (~300 ktpa) and 100% new-year-round process efficiency.
- **Escondida** – *Operator:* BHP (57.5%) with Rio Tinto, JOGMEC, IFC. *Location:* Antofagasta, Chile. *Type:* World's largest open-pit porphyry (producing since 1990s). *Geology:* Porphyry copper oxide and sulphide zones; ore ~1.0% Cu at surface. *Processing:* Solvent extraction/EW for oxide (cathode) plus milling/flotation for sulphides. *Risks:* Declining grades (now ~0.9–1.0% feed vs ~1.5% historically), water scarcity (delivers cathode via brine leaching), seismic events. *Mine Life:* Official guidance ~20+ years at current rates 【38†L658-L664】 . *Capex:* Sustained brownfields R&D (full-SaL leaching deployment, concentrator debottlenecks). *Ownership:* BHP majority; Rio/ others minority. *Cost:* 2025 unit cash cost \$1.19/lb 【38†L631-L634】 (\$2.62/kg), down 18% YOY on record 1.305 Mt output. By-product credits (some silver, gold, Mo) help but cost is still moderate on chart.
- **Collahuasi** – *Operator:* Glencore (44%) / Anglo (44%) / Japan (12%). *Location:* Tarapacá, Chile. *Type:* Huge open-pit porphyry. *Geology:* Large low-grade porphyry (~0.36% Cu feed 【37†L7-L10】). *Processing:* Mega-mill (60 Mtpa) plus flotation. *Risks:* Productivity affected by grade variability; challenging high-altitude logistics; water use. *Mine Life:* ~50 years. *Capex:* Ongoing expansion of water infrastructure; mechanization. *Ownership:* 2/3-ownership JV. *Cost:* Not separately reported, but Anglo's full-year cost of ~150¢/lb 【52†L113-L117】 (≈\$3.31/kg) for Collahuasi+Los Bronces reflects Collahuasi's influence; Collahuasi typically above \$1.50/lb cash cost.
- **Cerro Verde** – *Operator:* Freeport-McMoRan (54%) with Sumitomo. *Location:* Arequipa, Peru. *Type:* Open-pit porphyry. *Geology:* Moderately high-grade (~0.4% Cu) porphyry. *Processing:* Two milling/ flotation lines (copper concentrate, molybdenum by-product). *Risks:* Political/social (former owner controversies), power supply. *Mine Life:* >20 years. *Capex:* \$4B expansion (2015) doubling throughput; now only sustaining capex. *Ownership:* Majority Freeport. *Cost:* Freeport's 2025 consolidated cash cost was \$1.65/lb 【44†L61-L66】 (\$3.64/kg). Cerro Verde is one of FCX's lower-cost units due to scale and by-products (Mo, Ag), but we lack mine-specific AISC.
- **Buenaventura (Huarón/Otra Ha)** – *Operator:* Buenaventura (19%) and affiliates (owned 81% by Grana y Montero). *Location:* Pasco, Peru. *Type:* Underground polymetallic; includes Huarón (Cu-Zn-Ag-Pb) and Otra Ha (Cu-Zn) underground mines. *Geology:* Polymetallic veins and sediment-hosted stratabound. *Processing:* Flotation concentrator for copper, zinc, lead, silver. *Risks:* Reserve depletion; high inflation in Peru; royalties/taxes. *Mine Life:* ~5–10 years (reserves depleting). *Capex:* Minor (remnant projects). *Ownership:* Buenaventura 19%, rest public. *Cost:* Consolidated

with other mines, but Huarón reported C1 ~\$1.80/lb (2024) 【70†L198-L202】 (~\$3.97/kg) – higher than others due to smaller scale and complex ore.

- **KGHM (Poland – Lubin/Polkowice/Rudna)** – *Operator*: KGHM Polska Miedź. *Location*: Lower Silesia, Poland. *Type*: Large deep underground copper-polymetallic (Kupferberg porphyry and sediment-hosted). *Geology*: Porphyry copper with rich silver/gold. *Processing*: Crushing, milling, flotation to concentrate; then smelting/refining on-site. *Risks*: Aging infrastructure, geotechnical risk at depth, energy costs, Euro vs. USD exchange (złoty fluctuations). *Mine Life*: Several decades. *Capex*: €600M annual (new shafts, mechanization). *Ownership*: 100% state-owned. *Cost*: KGHM's FY2025 C1 was \$2.58/lb 【43†L55-L58】 (\$5.69/kg) – high on our list, due to high labor/energy in Poland. Substantial silver credits moderate net costs (KGHM's Q1 2026 reported C1 \$1.69 with massive silver credits 【42†L529-L537】 , but official FY35 cost was \$2.58).

Sensitivity Analysis

We modelled the impact of key factors on unit cost (for illustration, using an example mine with base cash cost \$/kg). A **fall in copper price** does not change cash cost per kg (which is largely technical), but it affects profit margin. For example, at a base price of \$10 000/t Cu (~\$4.54/lb), a low-cost mine at \$2.62/kg would earn \$4.54–2.62 = \$1.92/kg profit. If price drops 20% to \$8 000/t (\$3.64/lb), profit falls to \$1.02/kg.

By contrast, **energy costs** (fuel, power) and **grade** directly shift the cash cost. In our example, a 10% rise in diesel/electricity costs might raise cash costs by ~5–10% (assuming energy is a major input), e.g. from \$2.62 to ~\$2.85/kg. Similarly, a 10% drop in ore grade (less copper per tonne) effectively increases cost per kg by diluting output; e.g. going from 1.02% to 0.92% Cu feed would require ~10% more ore for the same metal, raising cash costs by ~10%. Figure below illustrates a stylized sensitivity: cash cost per kg vs. ±10–20% changes in energy cost or grade. (Note: actual sensitivities vary by mine – energy-intensive operations or lower grades will feel bigger swings.)

【figure】 *Figure: Sensitivity of cash cost (\$/kg) to input changes. A 10% rise in power/fuel or 10% drop in Cu grade significantly increases cost per kg, squeezing the margin. Copper price changes affect the net profit (not shown).*

Overall, even modest factor changes can shift low-cost mines above breakeven. High-current prices buffer profits today, but escalation in costs or grade declines could erode the cost advantage.

Data Gaps and Uncertainties

- **Cost Definitions**: Companies report various cost metrics (cash cost, C1, C2, AISC/C3) differently. We assume “cash cost” or “unit cost” refers to C1 (mine operating costs net of by-product credits). In several cases (e.g. Southern, Antofagasta), the figures are *net of by-products* 【57†L97-L100】 【72†L19-L23】 , which can swing with precious-metal prices. Where only gross costs were given (rarely), we omit them. Some firms include corporate overhead in AISC, others do not.
- **Currency Effects**: Producers in Chile, Peru and elsewhere incur costs in local currency (CLP, PEN) but report in USD. Exchange rate movements can affect costs. For example, Codelco's 2025 cost increase was partly due to a stronger CLP 【55†L17-L22】 . We used reported USD costs which already embed those effects.

- **Time-Frame:** We used the most recent full-year (2025 or latest fiscal year). Some mines reported only quarterly costs (e.g. Q1 Antofagasta 【72†L19-L23】). Where needed, we annualised or took year-end values. Early 2026 data (KGHM Q1) was used for 2025 when full-year 2025 wasn't yet published.
- **Production/Grade:** Exact annual production and grade per mine are often not publicly broken out. Our table uses best-known estimates or recent figures for 2024–25. In some cases (Buenavista, Collahuasi, Cerro Verde, others), production is aggregated or reported for entire company. We note these by footnotes and used available mining databases/press for context.
- **AISC:** Where we do quote AISC, it is often pro-forma (mines rarely publish mine-level AISC). For example, Southern's AISC was reported ~\$0.85/lb in 2025 (not shown above). Most entries above list only cash costs, as sustainable capital varies widely by mine and some estimates are not public.

Despite these uncertainties, the broad ranking is robust: Chilean porphyries like Buenavista, Oyu Tolgoi and Quellaveco operate at the very low end of the cost curve due to high grade and scale (and large by-product credits), whereas older complex mines (Codelco, KGHM, Teck's QB) sit at the high end. We caution that future changes (e.g. declining grades, rising diesel/prices, new regulations) could alter costs significantly.

Sources: Company disclosures (e.g. BHP FY2025 Operational Review 【38†L631-L634】 , Southern Copper FY2025 press release 【57†L97-L100】 , Freeport Q4/2025 report 【44†L61-L66】 , Antofagasta Q4/2025 report 【74†L199-L207】 , MMG 2025 interim results 【70†L198-L202】 , KGHM results 【43†L55-L58】 , etc.), and independent analyses 【55†L17-L22】 【52†L113-L117】 . All quotes are cited above.
